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INHERITANCE



The third tenet of OOP is *inheritance*, which is a mechanism for deriving a new class from an existing class. Rather than starting from scratch and potentially duplicating code, inheritance allows a programmer to write code for a new class that extends or differentiates it from an existing class.

Let's begin with a real-world example that demonstrates what inheritance is basically about. You're attending culinary school. One of your lessons involves an exhaustive demonstration of making hamburgers. You learn everything that there is to know about the different cuts of meat, the grinding of the meat, the best types of buns, the best lettuce, tomato, and condiments—just about everything you could imagine. You also learn about the best way to cook the hamburger, how long to cook it, when and how often to flip it over, and so on.

The next lesson in the curriculum is about cheeseburgers. The instructor *could* start from scratch and go through all the material about hamburgers again. But instead, they assume that you've retained the knowledge from the previous lesson and so already know everything there is to know about creating a great hamburger. This lesson therefore focuses on what types of cheese to use, when to add it, how much to use, and so on.